

Formatting	- o To reproduce this case study, follow the steps outlined below in order. - o The full analysis workflow includes exploratory data analysis (EDA), Welch's t-test, and Welch's ANOVA, with outputs generated at each stage for interpretation and submission. - o All scripts and data are available in the following Github: https://github.com/JulianneSweeten24/CS3
Purpose	• The purpose of this project is to demonstrate the ability to reproduce and interpret Welch's t-test and Welch's ANOVA using a real-world movie review dataset in Python. More specifically, the project investigates whether patterns in movie review length and sentiment severity provide evidence that aligns with the psychological concept of negativity bias: the tendency for negative experiences, emotions, or evaluations to have a stronger psychological impact than positive ones. By applying statistical testing methods to IMDB review data, the project examines whether negative reviews tend to differ in length from positive reviews and whether varying review length groups are associated with differences in sentiment extremity. By completing this project, the student should be able to: • Perform exploratory data analysis (EDA) on movie review data using a provided python file. • Understand how log transformation affects skewed review length distributions. • Reproduce and interpret Welch's t-test and Welch's ANOVA results in Python, including statistical significance and length group comparisons, using a provided python file. • Understand how RoBERTa sentiment severity scores are used in analysis. • Evaluate whether the findings align with negativity bias theory. • Interpret analytical outputs including plots, Welch's t-test results, and Welch's ANOVA results.
Notes	• For this project, our team used a RoBERTa model to calculate sentiment severity. Since replicating this is very time-consuming, we've provided the RoBERTa score dataset. (reviews_with_severity.csv). • OPTIONAL: We provided the roberta_sentiment.py file if you'd like to replicate the RoBERTa sentiment score calculation process.
Data	• All required data can be found within the repository's DATA folder.

	• The Appendix.pdf provides additional information about the dataset, as well as visualizations that support and inform the statistical reasoning used throughout the project (ie. the log transformation to create balanced length tertiles) • Download reviews_with_severity.csv and IMDB_Dataset.csv ◦ Store within an accessible folder on your computer.
EDA	• Download EDA_and_Cleaning_Modified.py from the SCRIPTS folder. ◦ Open the file within VSCode, Google Colab, or a similar environment, and run the script in its entirety. ■ Make sure the python file is in the <i>same folder</i> as IMDB_Dataset.csv. ■ If working with Google Colab, upload the IMDB_Dataset.csv to the environment. ◦ Running the full script will generate exploratory data analysis (EDA) visualizations, including IMDB review character count distributions, frequency distributions, a Q-Q plot, and visualizations demonstrating the effect of log transformation on the review length distribution. ◦ Save all generated plots and outputs for inclusion in the final submission.
Analysis	• Within the SCRIPTS folder, download the WelchANOVA.py file ◦ Open the file within VSCode, Google Colab, or a similar environment, and run the script in its entirety. ■ Make sure the python file is in the same folder as reviews_with_severity.csv ■ If working with Google Colab, upload the reviews_with_severity.csv to the environment. ◦ Running the full script will generate the statistical results for Welch's t-test and Welch's ANOVA. ◦ Save all generated outputs for inclusion in the final submission.
Output/Deliverable	• The output will include several visualizations generated during exploratory data analysis (EDA), along with the statistical results from Welch's t-test and Welch's ANOVA. • SUBMISSION DETAILS: Save or screenshot all relevant plots and statistical outputs, then compile them into a document where you answer the following questions. When answering, explicitly reference the relevant/corresponding visualization or statistical result that supports your interpretation. • Answer the following questions to guide understanding:

	○ How does applying a log transformation affect the normalization of the review length distribution? ○ Is the observed difference in review length between negative and positive reviews statistically significant? ○ Are the differences in sentiment extremity across the three review length groups statistically significant? ○ Do the results appear to align with negativity bias (the tendency for negative experiences or information to have a stronger psychological impact than positive ones)? Consider whether negative reviews tend to be longer or more emotionally extreme and whether the statistical results support this interpretation. ● Submit or share the completed document containing your plots, statistical outputs, and written interpretations.
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